CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

# Chapter 10: Circles

Every formula and result of the chapter on one sheet

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|---|---|---|
| 1 | Tangent and radius | Tangent $\perp$ radius at the point of contact |
| 2 | Tangents from an external point | $PA = PB$ |
| 3 | Length of a tangent | $PT = \sqrt{d^2 - r^2}$, where $d$ is the distance of the point from the centre |
| 4 | Distance of the point from the centre | $d = \sqrt{PT^2 + r^2}$ |
| 5 | Angle between two tangents | $\angle APB + \angle AOB = 180^\circ$ |
| 6 | Line from the point to the centre | $OP$ bisects $\angle APB$ and $\angle AOB$ |
| 7 | Number of tangents from a point | Inside: 0; on the circle: 1; outside: 2 |
| 8 | Quadrilateral circumscribing a circle | $AB + CD = AD + BC$ |
| 9 | Chord touching the inner of two concentric circles | Length $= 2\sqrt{R^2 - r^2}$ |
| 10 | Triangle with an incircle | $AF = AE = s - a$, $BD = BF = s - b$, $CD = CE = s - c$, where $s$ is the semi-perimeter |
| 11 | Radius of the incircle of a right-angled triangle | $r = \dfrac{a + b - c}{2}$, where $c$ is the hypotenuse |